

Kansas City's Bakery Trends: Exploring Opportunities for Colombian-Inspired Baked Goods

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Data Visualization Project

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Brief Description

Kansas City's bakery scene is diverse, however, ethnic baked goods, especially those from Colombia are underrepresented. Consumers are becoming more interested in unique, high-quality pastries, and this report aims to uncover opportunities to bring Colombian-inspired flavors to the market. By analyzing customer reviews, ratings, and bakery locations, I identify the most popular bakery items, evaluate how different bakery types perform, and determine the best locations for new bakery ventures. These insights will help business owners and investors make data-driven decisions on how to successfully introduce Colombian treats to Kansas City.

Business Benefit

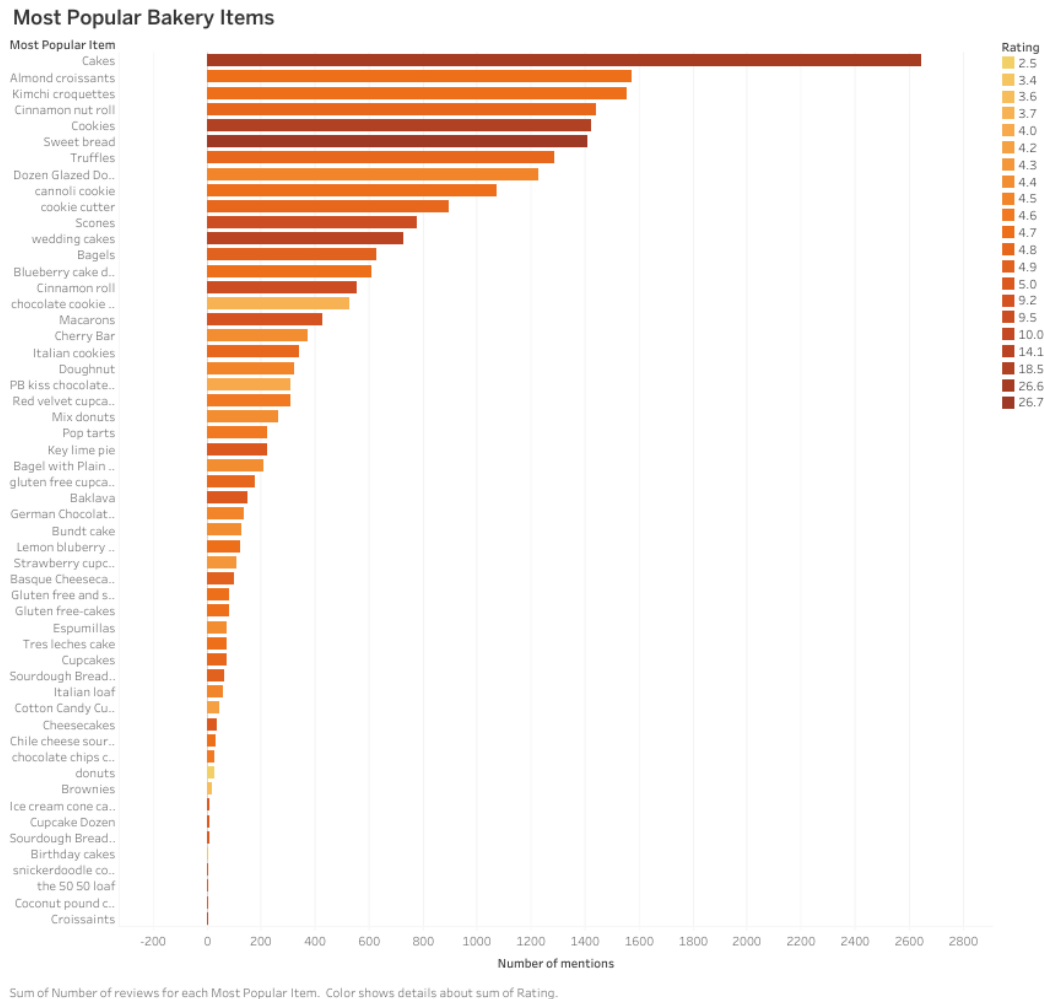
Understanding bakery trends is essential for business owners looking to expand or introduce new products. This study offers valuable insights into customer preferences, enabling entrepreneurs to identify potential opportunities for Colombian-inspired baked goods. Key Performance Indicators (KPIs) such as customer ratings, product popularity, and geographic distribution are used to measure satisfaction levels across different bakery types, identify the most frequently mentioned and highest-rated bakery items, and spot underserved areas in Kansas City that could serve as ideal locations for a new bakery. With these insights, bakery owners can bring Colombian-inspired baked goods to the market, meet customer demand, and fill a gap, helping to attract more customers and increase profits.

Data Visualizations and Analysis

Most popular bakery items

Identifying which bakery items are the most popular is essential for making strategic business decisions. The visualization below highlights the most frequently mentioned and highest-rated bakery items in Kansas City, showing clear preferences for certain treats. Cakes, almond croissants, and cinnamon nut rolls stand out as the most frequently mentioned and highly rated items. Additionally, specialty treats such as macarons, truffles, and tres leches cake also receive high customer approval, indicating a strong demand for unique, high-quality baked goods.

Figure 1: Most popular bakery items in Kansas City



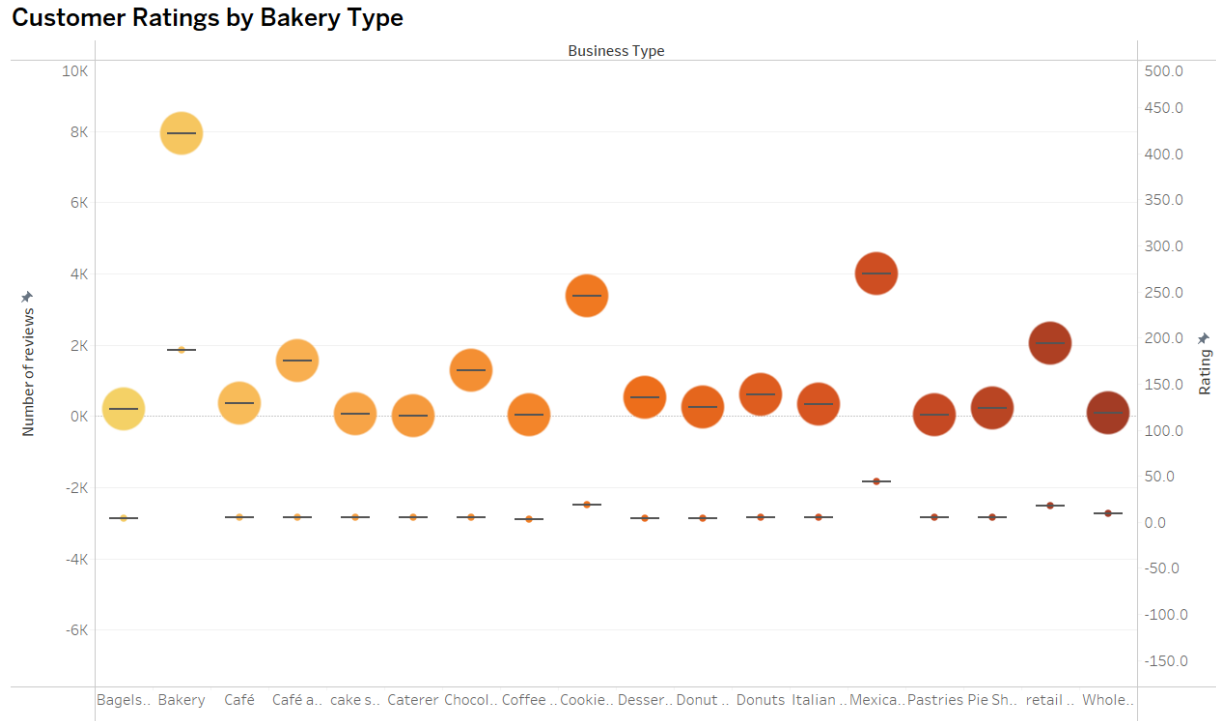
The rating distribution also shows that several frequently mentioned products maintain strong customer ratings. Notably, Latin-inspired pastries like tres leches cake and espumillas receive considerable attention, presenting an opportunity for growth in this category. However, Colombian treats such as pandebonos, pastelitos and empanadas remain underrepresented in local bakery offerings, revealing a potential gap in the market. Now that we understand which bakery items are most popular, let's explore how different types of bakeries perform based on customer ratings. This analysis will highlight whether customers prefer specialty bakeries over general bakeries and where Colombian-inspired pastries might have the strongest chance of success.

Customer ratings by bakery type

The dataset categorizes bakeries into general, ethnic, Latin-inspired, cafes, dessert shops, and specialty stores. The box plot below (fig. 2) reveals that ethnic and specialty bakeries receive higher ratings than general bakeries, indicating a preference for unique, high-quality, and culturally inspired baked goods in Kansas City. Latin-inspired bakeries show strong customer

satisfaction but are few in number, highlighting an underserved market for products like Colombian baked goods.

Figure 2: customer ratings by bakery type



Retail and traditional bakeries generate the most reviews, while donut shops and Mexican bakeries maintain competitive ratings, showing strong customer loyalty. Specialty cake shops and wholesale bakeries receive fewer reviews but sustain high ratings, likely due to their niche appeal. In contrast, chain bakeries accumulate more reviews but lower ratings, emphasizing that customers value quality over quantity.

While understanding bakery types and customer preferences is crucial, location also plays a key role in business success. A well-placed bakery can attract more foot traffic and build a loyal customer base. Let's examine the geographic distribution of bakeries in Kansas City to identify underserved areas with high potential for a Colombian bakery.

Geographic distribution of bakeries

The map visualization below (fig.3.1 and fig. 3.2) illustrates the distribution of bakeries across Kansas City, highlighting both bakery-dense areas and potential market gaps. Downtown and midtown have a high concentration of bakeries, meaning new businesses in these areas would face stronger competition. In contrast, certain neighborhoods, particularly in South Kansas City and some suburban areas, have fewer bakeries, presenting potential opportunities for new ventures. Additionally, the most highly rated bakeries tend to be clustered in specific areas,

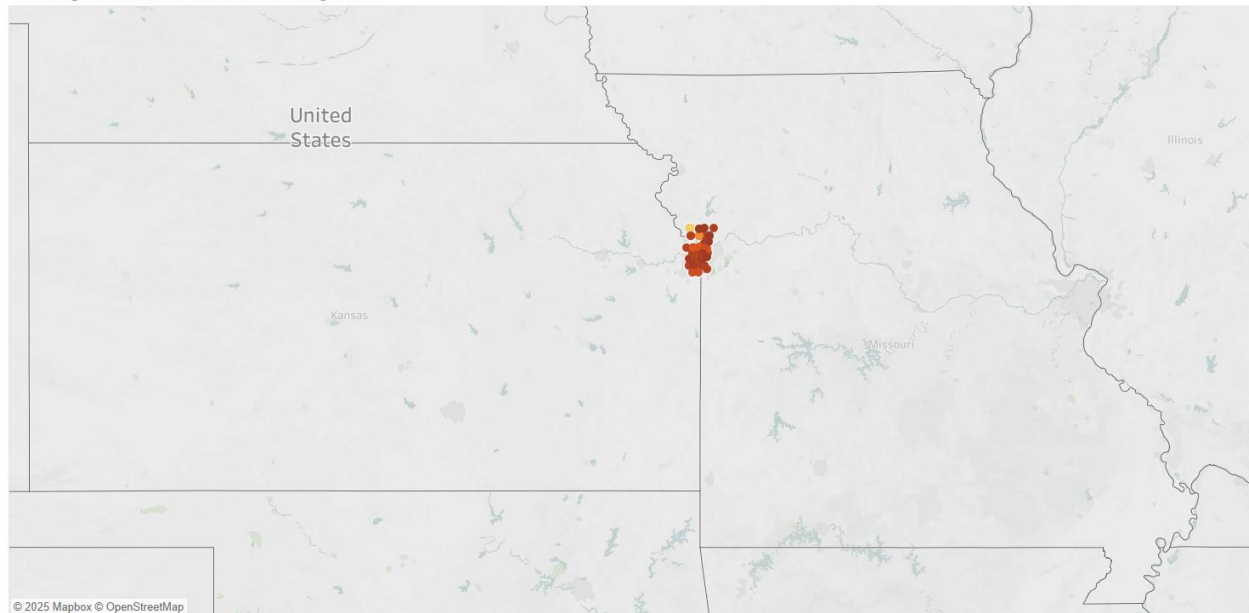
suggesting that location influences customer satisfaction, either through high foot traffic or strong community loyalty. For investors, these insights can help guide location-based decisions when opening a Colombian-inspired bakery. Selecting an area with less competition but strong demand could maximize profitability and brand impact.

Figure 3.1: Geographic distribution of bakeries (close view)

Bakery Locatons in Kansas City Area



Figure 3.2: Geographic distribution of bakeries (states view)

Bakery Locations in Kansas City Area

Conclusion

The data tells a clear story: Kansas City's bakery market is changing and evolving, with more people actively looking for unique, high-quality baked goods. Colombian pastries are still uncommon in the area, which creates a golden opportunity for a bakery that combines traditional and authentic flavors with local preferences. By strategically choosing a location with low competition, focusing on high-quality ingredients, and meeting customer interest in specialty products, a Colombian bakery can not only succeed but also introduce a rich culinary experience to the community. The next step for this project? Turning these insights into action and bringing a taste of Colombia to Kansas City's food scene.

Access to work materials

Now let's discuss our sources, the data and visualizations for this project were created by using a combination of the following tools and methods:

- **For data collection:** I manually collected bakery reviews, number of reviews, ratings, and full addresses from Google Maps. Additionally, I downloaded these two open datasets from KCMO Open Data: KCMO Food Permit 2025 and Business License Holders 2025
- **For data processing:** I used Python to merge the two datasets from KCMO Open Data and then saved the result as an Excel file named `Cleaned_Data_Bakeries`. In that same file, I manually input the information collected from Google Maps, such as reviews, ratings, number of reviews, and complete addresses.
- **For data cleaning and entry:** The `Cleaned_Data_Bakeries.xlsx` file served as the primary dataset for this project, containing both the merged open data and the manually collected Google Maps data.
- **For data analysis:** I created a separate Python script to run some initial analysis of the dataset before moving on to visualizations, this helped me understand the structure of the data and clean any inconsistencies.
- **Finally, for data visualization:** I used Tableau to build the visualizations and dashboard that show the findings from my analysis. The visualizations include insights into popular bakery items, customer ratings by bakery type, and geographic distribution of bakeries. All dashboard and visualizations are compiled in a Tableau workbook (.twbx).

The project materials, including the cleaned dataset, Python analysis files, and Tableau workbook, are available in the accompanying GitHub repository.

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